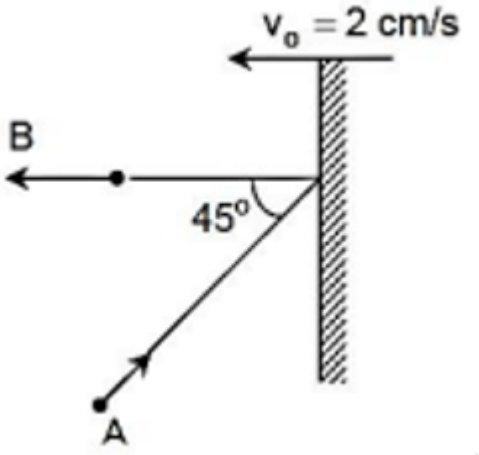
Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

Mirror is moving towards the particle with speed $2\ \text{cm/sec}$ speed of A and B are $10\sqrt{2}$ and $5\ \text{cm/sec}$ respectively in the direction as shown in figure. Magnitude of velocity of image of the particle B with respect to image of A.



(A) $\sqrt{325}\ \text{cm/sec}$

(B) $15\ \text{cm/sec}$

(C) $13\ \text{cm/sec}$

(D) $\sqrt{269}\ \text{cm/sec}$

Q.2

The plane surface of a planoconvex lens is silvered. If radius of curved surface is R and refractive index is μ, then the system behaves like a concave mirror whose radius will be

(A) $\frac{\text{R}}{\mu}$

(B) $\text{R}\mu$

(C) $\frac{\text{R}}{\mu-1}$

(D) $R(\mu-1)$

Q.3

A prism of dispersive power 0.021 and refractive index 1.53 form an achromatic combination with prism of angle 4.2° and dispersive power 0.045 having refractive index 1.65. Find the resultant deviation.

(A) 1.12°

(B) 2.16°

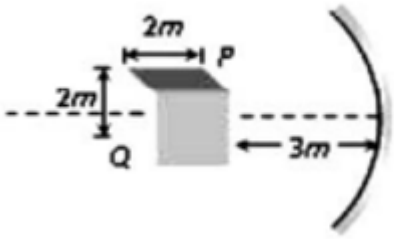
(C) 3.12°

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(D) 4.18°

Q.4

A cube of side $2m$ is placed in front of a concave mirror focal length $1\ \text{m}$ with its face P at a distance of $3\ \text{m}$ and face Q at a distance of $5\ \text{m}$ from the mirror. The distance between the images of face P and Q and height of images of P and Q are



(A) $1\ \text{m}, 0.5\ \text{m}, 0.25\ \text{m}$

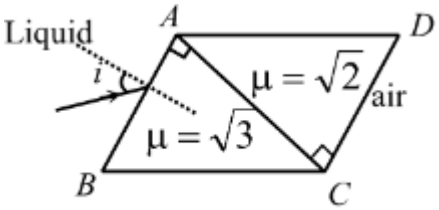
(B) $0.5\ \text{m}, 1\ \text{m}, 0.25\ \text{m}$

(C) $0.5\ \text{m}, 0.25\ \text{m}, 1\ \text{m}$

(D) $0.25\ \text{m}, 1\ \text{m}, 0.5\ \text{m}$

Q.5

A ray of light from a liquid $(\mu = \sqrt{3})$ is incident on a system of two right angled prism of refractive indices $\sqrt{3}$ and $\sqrt{2}$ as shown. The ray suffers zero deviation when emerges into air from CD. The angle of incidence i is



(A) 45°

(B) 35°

(C) 20°

(D) 10°

Q.6

Determine the angle of flint glass prism, which should be combined with a crown glass prism of 5° so as to give dispersion but no deviation. Given, for crown glass : $\mu_\text{v} = 1.523, \mu_\text{r} = 1.515$; for flint glass: $\mu'_\text{v} = 1.688, \mu'_\text{r} = 1.650$.

(A) -1.14°

(B) -3.88°

(C) -2.12°

(D) -4.12°

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Q.7

The dispersive powers of crown and flint glasses are 0.03 and 0.05 respectively. The refractive indices for yellow light for these glasses are 1.517 and 1.621 respectively . It is desired to form an achromatic combination of prisms of crown and flint glasses which can produce a deviation of 1° in the yellow ray. Find the refracting angles of the two prisms needed.

(A) 2.4°

(B) 1.4°

(C) 3.4°

(D) 5.2°

Q.8

A point object is placed on the principal axis of a concave mirror, at a distance of 15 cm from the pole. The radius of curvature of the mirror is 20 cm and the object is made to oscillate along the principal axis with an amplitude of 2 mm. The amplitude of its image will be

(A) 2 mm

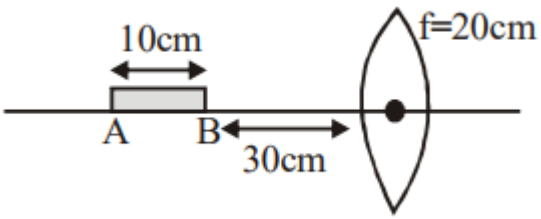
(B) 4 mm

(C) 8 mm

(D) 16 mm

Q.9

AB is linear object placed along optical axis as shown in figure. Tick the incorrect statement(s) -



(A) The length of image is smaller than the length of object

(B) The length of image is larger than the length of object

(C) The length of image is equal to the length of object

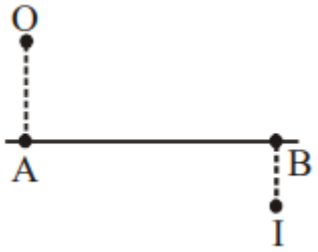
(D) If middle portion of the lens is painted then the length of image is smaller than length of object.

Q.10

A luminous point object is placed at O, whose image is formed at I as shown in figure. Line AB is the optical axis.

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Which of the following statement is/are correct -



(A) If a lens is used to obtain the image, then it must be a converging lens and its optical centre will be the intersection point of line AB and OI.

(B) If a lens is used to obtain the image, then it must be a diverging lens and its optical centre will be the intersection point of line AB and OI.

(C) If a mirror is used to obtain the image, then the mirror must be concave and object and image subtend equal angles at the pole of the mirror.

(D) I is real image.

Q.11

Which of the following statements is/are correct about the refraction of light from a plane surface when light ray is incident in denser medium. (C is critical angle)

(A) The maximum angle of deviation of during refraction is $\frac{\pi}{2} - C$, it will be at angle of incidence is C.

(B) The maximum angle of deviation for all angle incidences is $\pi - 2C$, when angle of incidence is slightly greater than C.

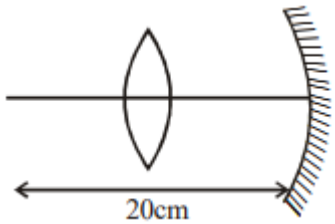
(C) If angle of incidence is less than C then deviation increases if angle of incidence is also increased.

(D) If angle of incidence is greater than C then angle of deviation decreases if angle of incidence is increased.

Q.12

Passage

A concave mirror of radius of curvature $20\ cm$. is shown in the figure. A circular disc of diameter $(1.0 + 0.2t)cm$. is placed on the principle axis of mirror with its plane perpendicular to the principle axis at a distance $15\ cm$. from the pole of the mirror.



Question

The image formed by the mirror will be in the shape of a -

(A) Circular disc

(B) Elliptical disc with major axis horizontal

(C) Elliptical disc with major axis vertical

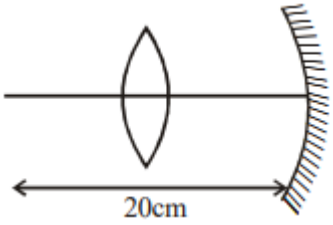
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(D) Distorted disc

Q.13

Passage

A concave mirror of radius of curvature $20\ \text{cm}$. is shown in the figure. A circular disc of diameter $(1.0 + 0.2t)\text{cm}$. is placed on the principle axis of mirror with its plane perpendicular to the principle axis at a distance $15\ \text{cm}$. from the pole of the mirror.



Question

In the above question, the area of the image of the disc at $t = 1$ second is $-$

(A) $1.2\pi\text{cm}^2$

(B) $1.44\pi\text{cm}^2$

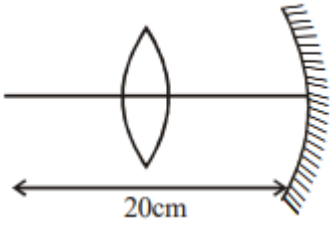
(C) $1.52\pi\text{cm}^2$

(D) None of these

Q.14

Passage

A concave mirror of radius of curvature $20\ \text{cm}$. is shown in the figure. A circular disc of diameter $(1.0 + 0.2t)\text{cm}$. is placed on the principle axis of mirror with its plane perpendicular to the principle axis at a distance $15\ \text{cm}$. from the pole of the mirror.



Question

What will be the rate at which the horizontal radius of image will be changing -

(A) $0.2\ \text{cm/s}$ increasing

(B) $0.2\ \text{cm/s}$ decreasing

(C) $0.4\ \text{cm/s}$ increasing

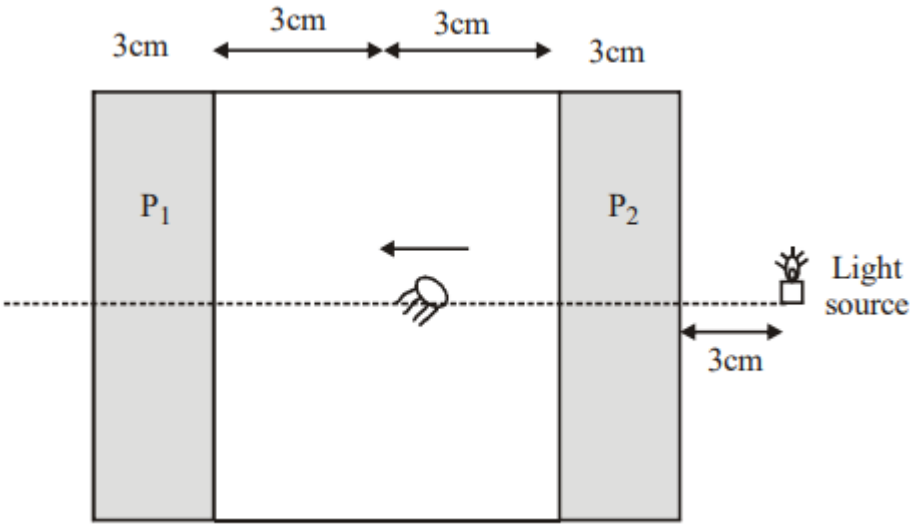
(D) $0.4\ \text{cm/s}$ decreasing

Q.15

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Passage

There is an insect inside a cabin eying towards a thick glass plate P_1. Insect sees the images of light source across the glass plate P_1 outside the cabin. Cabin is made of thick glass plates of refractive index $\mu = 3/2$ and thickness $3\ \text{cm}$. Insect is eying from the middle of the cabin as shown in figure. (Glass plates are partially reflective and consider only paraxial rays)



Question

At what distance (from eye of insect) will the eye see first image ?

(A) $5\ \text{cm}$.

(B) $7\ \text{cm}$.

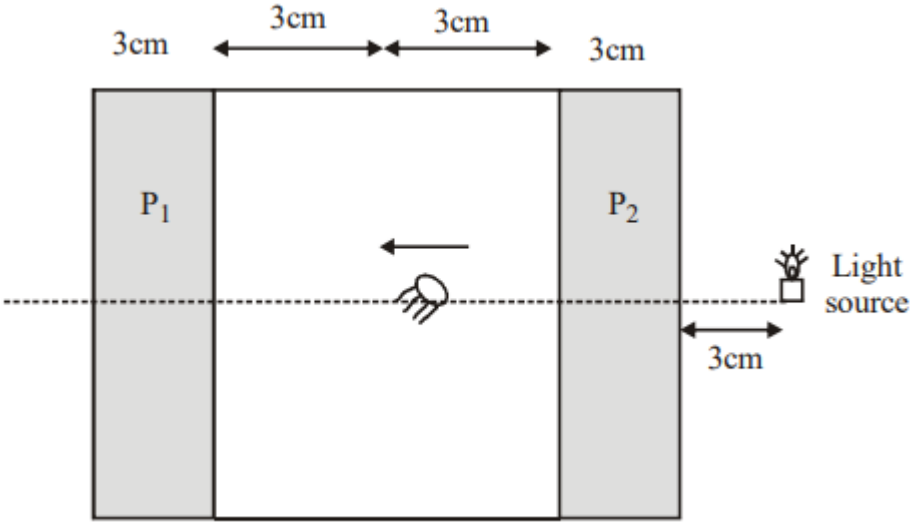
(C) $9\ \text{cm}$.

(D) $14\ \text{cm}$.

Q.16

Passage

There is an insect inside a cabin eying towards a thick glass plate P_1. Insect sees the images of light source across the glass plate P_1 outside the cabin. Cabin is made of thick glass plates of refractive index $\mu = 3/2$ and thickness $3\ \text{cm}$. Insect is eying from the middle of the cabin as shown in figure. (Glass plates are partially reflective and consider only paraxial rays)



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Question

At what distance (from eye of insect) will the eye see second image ?

(A) 11 cm.

(B) 13 cm.

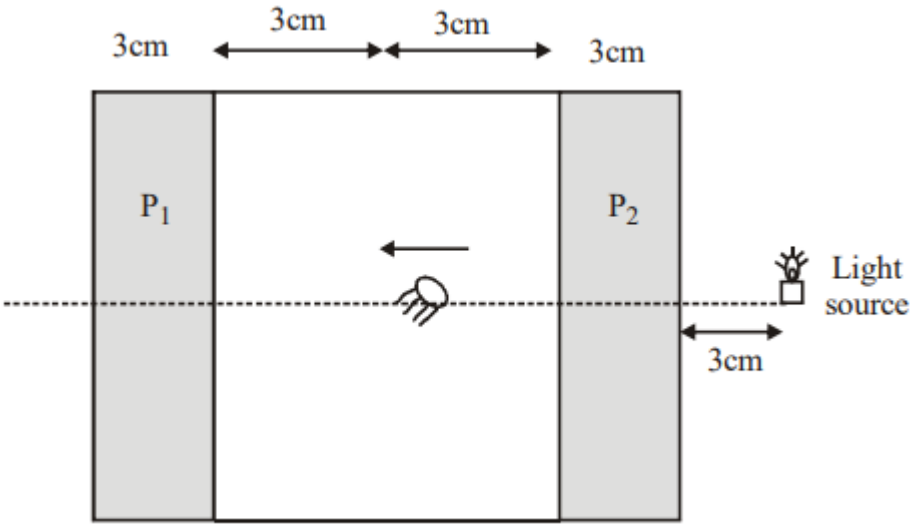
(C) 16.5 cm.

(D) 18 cm.

Q.17

Passage

There is an insect inside a cabin eying towards a thick glass plate P_1. Insect sees the images of light source across the glass plate P_1 outside the cabin. Cabin is made of thick glass plates of refractive index $\mu = 3/2$ and thickness 3 cm. Insect is eying from the middle of the cabin as shown in figure. (Glass plates are partially reflective and consider only paraxial rays)



Question

Number of image seen by insect?

(A) 2

(B) 4

(C) 8

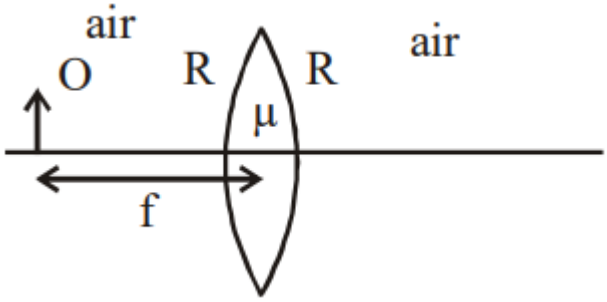
(D) ∞

Q.18

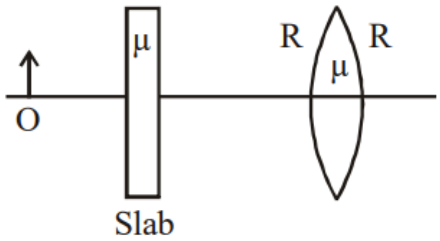
An object O (real) is placed at focus of an equi-convex lens as shown in figure. The refractive index of lens is $\mu = 1.5$ and the radius of the curvature of either surface of lens is R. The lens is surrounded by air. In each statement of column-I some changes are made to situation given above and information regarding final image formed as a result is given in column-II. The distance between lens and object is unchanged in all statements of column-I. Match the

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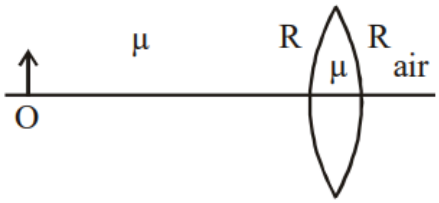
statements in column-I with resulting image in column-II.



Column I	**Column II**
(A) If the refractive index of the lens is doubled (that is, made 2μ) then	(p) final image is real
(B) If the radius of curvature is doubled (that is, made 2R) then	(q) final image is virtual
(C) If a glass slab of refractive index $\mu = 1.5$ is introduced to size of image before the change was made.	(r) final image becomes smaller in size in comparison between the object and lens as shown, then



(D) If the left side of lens is filled with a medium of refractive index $\mu = 1.5$, as shown, then	(s) final image is of same size of object.



(A) $(C) - p, (B) - r, q, (C) - s, r, (D) - q$

(B) $(A) - s, p(B) - p, (C) - s, (D) - q$

(C) $(A) - q, r(B) - q, r(C) - p, q, r, s(D) - p, s$

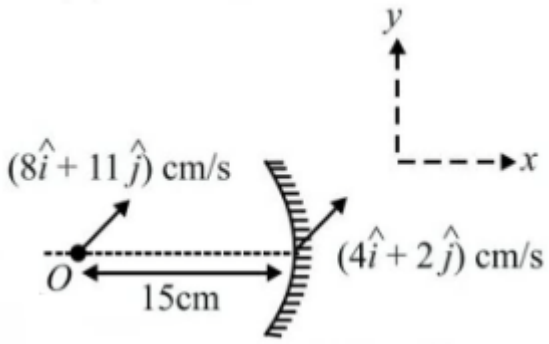
(D) $(A) - p, r, (B) - q, r, (C) - q, r, (D) - q, r$

Q.19

A point object is located at a distance $15\ \text{cm}$ from the pole of a concave mirror of focal length $10\ \textbf{cm}$ on its principal axis is moving with a velocity $(8\hat{i} + 11\hat{j})\textbf{cm/s}$ and velocity of mirror is $(4\hat{i} + 2\hat{j})\text{cm/s}$ as shown. If

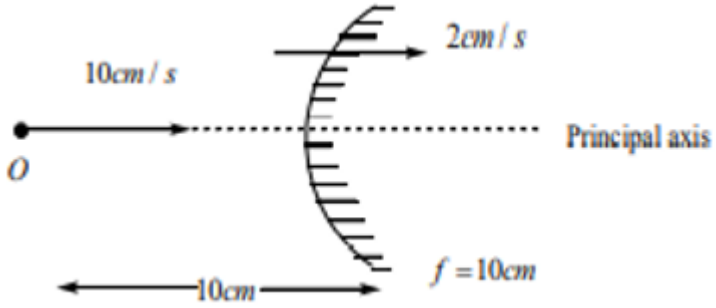
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$\vec{V}$ is the velocity of image. Then find the value of $\left|\frac{V}{5}\right|$ (in $\mathrm{cm/s}$) at the instant shown in figure



Q.20

What is the velocity (in $\mathrm{cm/s}$) of image in situation shown below ($O = 0bject$, f = focal length). Object moves with velocity $10\ \mathrm{cm/s}$ and mirror moves with velocity $2\ \mathrm{cm/s}$ as shown



Answers & Solutions

Q.1 Answer:

—

Solution:

Velocity of image of particle B,

$\overset{u}{V}_B = 5i + 4(-i) = i$

Velcoity of image of particle A,

$\overset{u}{V}_A = 10(-i) + 4(-i) - 10j$

$\therefore \overset{ur}{V}_A = -14i - 10j$

Relative velocity of image

$\overset{u}{V}_{BA} = \overset{u}{V}_B - \overset{u}{V}_A$

$= I - [14(-i) - 10j]$

$= 15i + 10j$

$\therefore \overset{ur}{V}_{BA} = \sqrt{325}\ \mathrm{cm/sec}$

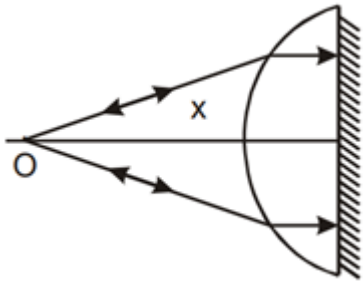
Q.2 Answer:

$\frac{\mathrm{R}}{\mu - 1}$

Solution:

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Let the image of an object at O is formed at the same point as shown in figure. The distance of O from the plane surface is x. The rays suffer refraction at first surface (curved) as they reach lens. After wards become parallel and gets reflected form plane surface and so retrace the path and image is formed at O itself.



$$\frac{\mu}{v}-\frac{1}{u}=\frac{\mu-1}{R}$$
$$u=-x, v=\infty$$
$$\frac{\mu}{\infty}+\frac{1}{x}=\frac{\mu-1}{R}$$
$$x=\frac{R}{\mu-1}$$

As such O behaves as equivalent to centre of curvature of equivalent concave mirror.

$$\therefore \quad \text{Radius } = x = \frac{R}{\mu - 1}$$

Q.3 Answer:

3.12°

Solution:

Here, $\omega = 0.021; \mu = 1.53; \omega' = 0.045;$

$$\mu' = 1.65; \quad A' = 4.2^\circ$$

For no dispersion, $\omega\delta + \omega'\delta' = 0$ **or** $\omega A(\mu - 1) + \omega' A'(\mu - 1) = 0$

or

$A = -\frac{\omega' A'(\mu' - 1)}{\omega(\mu - 1)} = -\frac{0.045 \times 4.2 \times (1.65 - 1)}{0.021 \times (1.53 - 1)} = -11.04^\circ$

Net deviation, $\delta + \delta' = A(\mu - 1) + A'(\mu' - 1)$

$= -11.04(1.53 - 1) + 4.2(1.65 - 1) = -11.04 \times 0.53 + 4.2 \times 0.65 = -5.85 + 2.73 = 3.12^\circ$

Q.4 Answer:

$0.25 \text{ m}, 1 \text{ m}, 0.5 \text{ m}$

Solution:

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For surface P, $\frac{1}{v_1} = \frac{1}{f} - \frac{1}{u} = 1 - \frac{1}{3} = \frac{2}{3} \Rightarrow v_1 = \frac{3}{2}m$

For surface Q, $\frac{1}{v_2} = \frac{1}{f} - \frac{1}{u} = 1 - \frac{1}{5} = \frac{4}{5} \Rightarrow v_2 = \frac{5}{4}m$

$\therefore v_1 - v_2 = 0.25m$

Magnification of P $= \frac{v_1}{u} = \frac{3/2}{3} = \frac{1}{2}$

$\therefore$ **Height of P** $= \frac{1}{2} \times 2 = 1m$

Magnification of Q $= \frac{v_2}{u} = \frac{5/4}{5} = \frac{1}{4}$

$\therefore$ **Height of Q** $= \frac{1}{4} \times 2 = 0.5m$

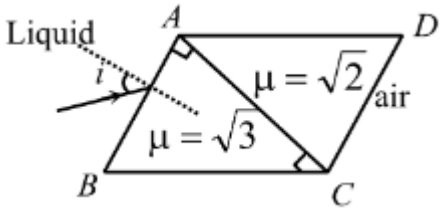
Q.5 Answer:

45°

Solution:

$\sqrt{3}\sin(90 - i) = \sqrt{2}\sin r$

$1\sin i = \sqrt{2}\sin(90^\circ - r)$



on solving we get $i = 45^\circ$

$\therefore \quad$ (a)

Q.6 Answer:

-3.88°

Solution:

Crown glass, $\mu_V = 1.523; \mu_r = 1.515; A = 5^\circ$

Flint glass $\mu_v' = 1.688; \mu_r' = 11.650; A' = ?$

So that the combination of the two prisms does not produce any deviation,

$$A' = -\frac{A(\mu - 1)}{\mu' - 1}$$

$$\text{Now, } \mu = \frac{\mu_V + \mu_r}{2} = \frac{1.523 + 1.515}{2} = 1.519 \text{ and } \mu' = \frac{\mu_V' + \mu_r'}{2} = \frac{1.688 + 1.650}{2} = 1.669$$

$$\therefore \quad A' = -\frac{5(1.519 - 1)}{1.669 - 1} = -\frac{5 \times 0.519}{0.669} = -3.88^\circ$$

Q.7 Answer:

2.4°

Solution:

Suppose, the angle of the crown prism needed is A and that of the flint prism is A'.

We have $\omega = \frac{\mu_v - \mu_r}{\mu - 1}$ **or,** $\mu_v - \mu_r = (\mu - 1)\omega$

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The angular dispersion produced by the crown prism is $(\mu_v - \mu_r)\,A = (\mu - 1)\omega A$

Similarly, the angular dispersion produced by the flint prism is $(\mu' - 1)\,\omega' A'$

For achromatic combination, the net dispersion should be zero. Thus,

$$(\mu - 1)\omega A = (\mu' - 1)\,\omega' A'$$
$$\text{or} \quad \frac{A'}{A} = \frac{(\mu-1)\omega}{(\mu'-1)\omega'} = \frac{0.517\times0.03}{0.621\times0.05} = 0.50 \quad .....(1)$$

The deviation in the yellow ray produced by the crown prism is $\delta = (\mu - 1)\,A$ **and by the flint prism is** $\delta' = (\mu' - 1)\,A'$**. The net deviation produced by the combination is**

$$\delta - \delta' = (\mu - 1)A - (\mu' - 1)\,A'$$
$$\text{or} \quad 1^\circ = 0.517\,A - 0.621\,A' \quad .....(2)$$

Solving (1) and (2), $A = 4.8^\circ$ **and** $A' = 2.4^\circ$**. Thus, the crown prism should have its refracting angle** 4.8° **and that of the flint prism should be** 2.4°**.**

Q.8 Answer:

8 mm

Solution:

Using mirror formula,

$$\frac{1}{-10} = \frac{1}{v} + \frac{1}{-15} \Rightarrow v = -30 \text{ cm}$$

$$|\text{Axial magnification}| = \frac{V^2}{u^2} = \left(\frac{30}{15}\right)^2 = 4$$

Amplitude of image = $4 \times 2 = 8$ **mm.**

Q.9 Answer:

The length of image is smaller than the length of object

Solution:

For image of B, $\frac{1}{v_B} - \frac{1}{-30} = \frac{1}{20} \Rightarrow v_B = 60$ cm
For image of A, $\frac{1}{v_A} - \frac{1}{-40} = \frac{1}{20} \Rightarrow v_A = 40$ cm.
Length of image $= v_B - v_A = 20$ cm.

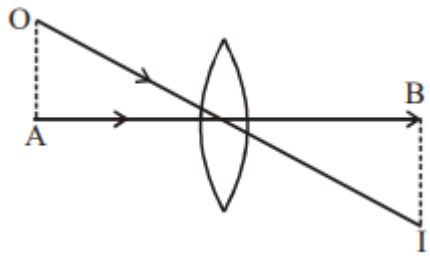
Q.10 Answer:

If a lens is used to obtain the image, then it must be a converging lens and its optical centre will be the intersection point of line AB and OI.

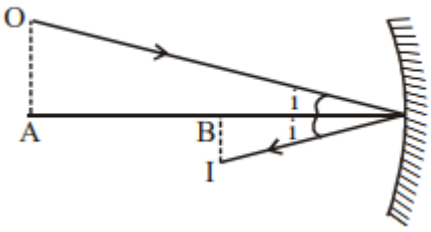
Solution:

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(1)



(3)

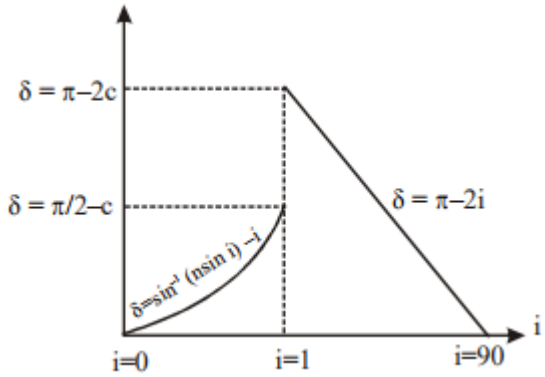


(4) Image is inverted $\Rightarrow$ It should be real

Q.11 Answer:

The maximum angle of deviation of during refraction is $\frac{\pi}{2} - \mathrm{C}$, it will be at angle of incidence is C.

Solution:



When angle of incidence is equal to the critical angle, deviation $= \frac{\pi}{2} - \mathrm{C}$

When angle of incidence is greater than C, total deviation is $\pi - 2\mathrm{i}$ (decreases as i increases), which is greatest for $\mathrm{i} = \mathrm{C}$, thus maximum deviation $= \pi - 2\mathrm{C}$

When angle of incidence is less than C, deviation is $r - i = \sin^{-1}(\mu \sin i) - i$, which increases as i increases.

Q.12 Answer:

Circular disc

Solution:

$$\frac{h_i}{h_0} = \frac{-v}{u} = \frac{r_i}{r_0} = \frac{-v}{u} = \frac{-\frac{uf}{u-f}}{u} = \frac{-f}{u-f}$$

So only dimension of image change not shape.

(v is same for every part)

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Q.13 Answer:

$1.2\pi \text{cm}^2$

Solution:

$$\frac{r_i}{r_0} = \frac{-v}{u} \Rightarrow r_i = -r_0\frac{v}{u} = -\frac{r_0 f}{u-f} = 1.2 \text{ cm}$$

Q.14 Answer:

0.2 cm/s increasing

Solution:

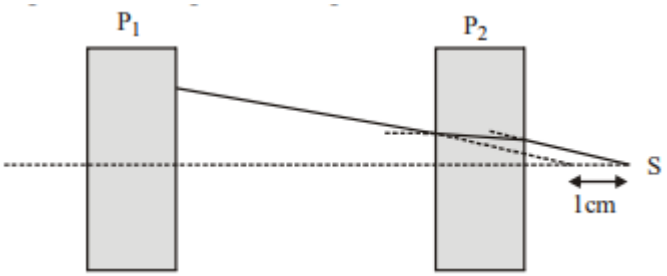
$$\frac{dr_i}{dt} = +0.2$$

Q.15 Answer:

$14 \text{ cm}.$

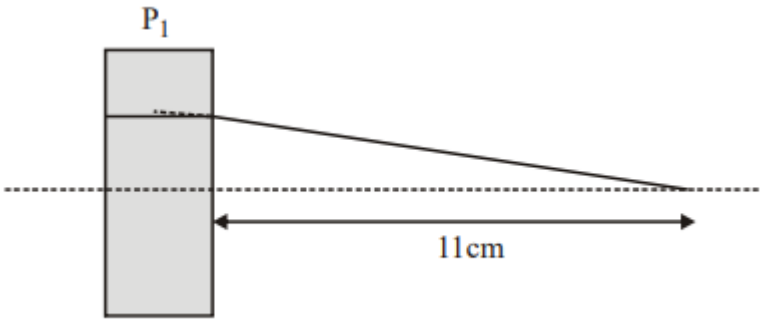
Solution:

(i) As it is possible only when, when light reflects from P_1 after refraction from the plate P_2 of light coming from light source.



First image will form due to reflection from the right surface of P_1. As light ray is falling on P_1 from 11 cm. so it will form image at 11 cm. in left. So, distance of first image from insect is $11 + 3 = 14 \text{ cm}$.

(ii) Second image will form due to reflection on left surface of P_1.



For left surface of P_1 light seems to coming from the distance $= \frac{33}{2} + 3 = \frac{39}{2}$ cm.

So, light seems to coming from $\frac{45}{2}$ cm. from right surface of P_1.

So, final position of second image will be $\frac{45/2}{3/2} = 15$ cm.

So, distance of second image from insect $= 18$ cm.

(iii) Due to multiple reflections infinite image will be formed.

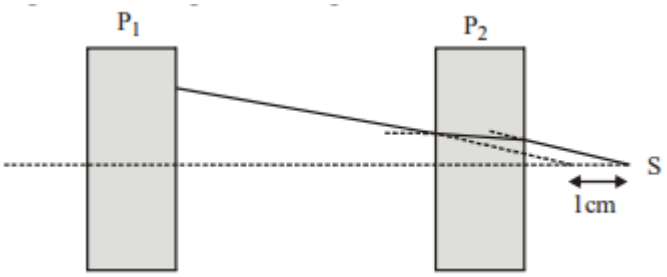
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Q.16 Answer:

18 cm.

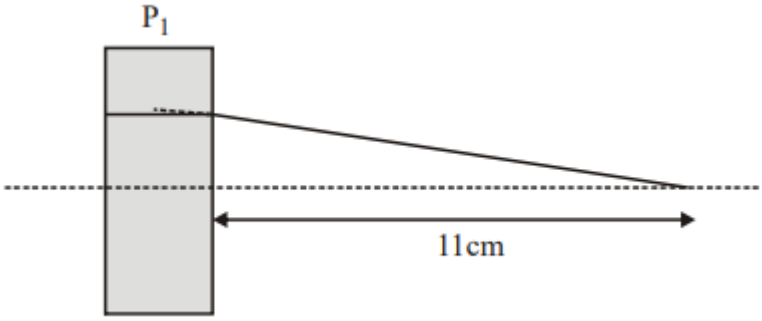
Solution:

(i) As it is possible only when, when light reflects from P_1 after refraction from the plate P_2 of light coming from light source.



First image will form due to reflection from the right surface of P_1. As light ray is falling on P_1 from 11 cm. so it will form image at 11 cm. in left. So, distance of first image from insect is $11 + 3 = 14$ cm .

(ii) Second image will form due to reflection on left surface of P_1.



For left surface of P_1 light seems to coming from the distance $= \frac{33}{2} + 3 = \frac{39}{2}$ cm.

So, light seems to coming from $\frac{45}{2}$ cm. from right surface of P_1.

So, final position of second image will be $\frac{45/2}{3/2} = 15$ cm.

So, distance of second image from insect $= 18$ cm.

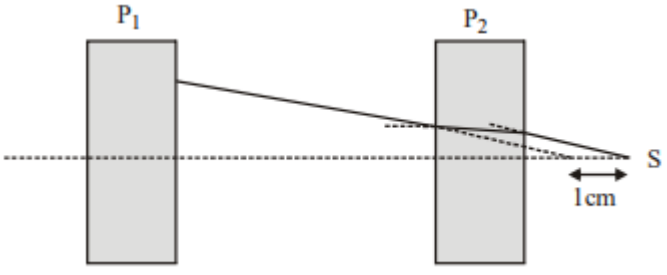
(iii) Due to multiple reflections infinite image will be formed.

Q.17 Answer:

∞

Solution:

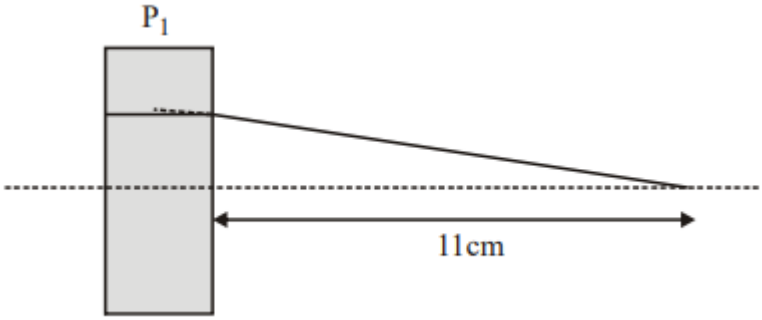
(i) As it is possible only when, when light reflects from P_1 after refraction from the plate P_2 of light coming from light source.



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First image will form due to reflection from the right surface of P_1. As light ray is falling on P_1 from 11 cm. so it will form image at 11 cm. in left. So, distance of first image from insect is $11 + 3 = 14$ cm .

(ii) Second image will form due to reflection on left surface of P_1.



For left surface of P_1 light seems to coming from the distance $= \frac{33}{2} + 3 = \frac{39}{2}$ cm.

So, light seems to coming from $\frac{45}{2}$ cm. from right surface of P_1.

So, final position of second image will be $\frac{45/2}{3/2} = 15$ cm.

So, distance of second image from insect $= 18$ cm.

(iii) Due to multiple reflections infinite image will be formed.

Q.18 Answer:

$(\mathrm{A}) - \mathrm{p}, \mathrm{r}, (\mathrm{B}) - \mathrm{q}, \mathrm{r}, (\mathrm{C}) - \mathrm{q}, \mathrm{r}, (\mathrm{D}) - \mathrm{q}, \mathrm{r}$

Solution:

$(A) - pr, (B) - qr, (C) - qr, (D) - qr.$

Initially the image is formed at infinity.

(A) As μ is increased the focal length decreases. Hence the object is at a distance larger than focal length. Therefore final image is real. Also final image becomes smaller in size in comparison to size of image before the change was made.

(B) If the radius of curvature is doubled, the focal length decreases. Hence the object is at a distance lesser than focal length. Therefore final image is virtual. Also final image becomes smaller in size in comparison to size of image before the change was made.

(C) Due to insertion of slab the effective object for lens shifts right wards. Hence final image is virtual. Also final image becomes smaller in size in comparison to size of image before the change was made.

(D) The object comes to centre of curvature of right spherical surface as a result. Hence the final image is virtual. Also final image becomes smaller in size in comparison to size of image before the change was made.

Q.19 Answer:

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Solution:

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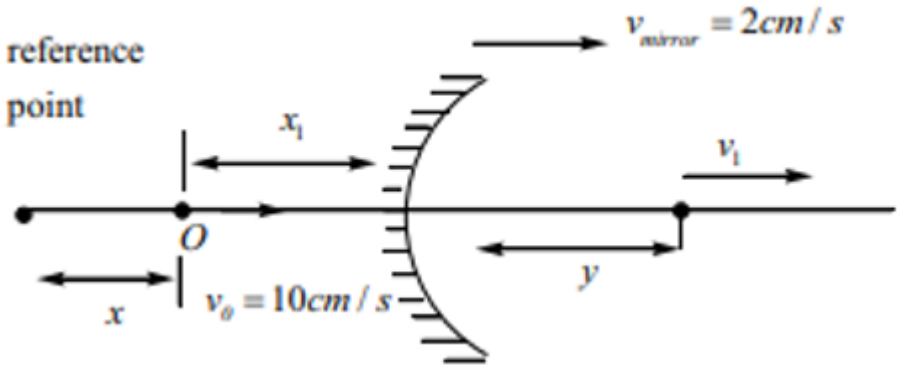
$\frac{dx}{dt} = 4, \frac{dy}{dt} = 9$
$u = -x$
$-\frac{1}{10} = \frac{1}{V} + \frac{1}{-x}$
$V = \frac{-10x}{x-10}$
$V_{1x} = \frac{dV}{dt} = \left(\frac{10}{x-10}\right)^2 \left(\frac{dx}{dt}\right)$
$V_{1,} = -16$
$m = -\frac{V}{-x} = \frac{-10}{x-10} = \frac{y_1^0}{y}$
$V_{1y} = -\left(\frac{10}{x-10}\right) \frac{dy}{dt}$
$V_{1y} = -18$
$V_{\text{image}} = -16\hat{i} - 18\hat{j}$
$V_{\text{image w.r.t mirror}} = -12\hat{i} - 16\hat{j}$

Q.20 Answer:

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Solution:

$$\frac{1}{y} + \frac{1}{-x_1} = \frac{1}{10}$$
$$\frac{1}{y} - \frac{1}{10} = \frac{1}{10} \Rightarrow y = 5 \text{ cm}$$



$$\frac{dx}{dt} = 10 \text{ cm/s}, \frac{d(x + x_1)}{dt} = 2 \text{ cm/s}$$
$$\frac{dy}{dt} = v_1 = 2 \text{ cm/s}$$
$$\frac{dy}{dt} = \left(\frac{y^2}{x_2}\right) \frac{dx_1}{dt} \Rightarrow v_1 - 2 = \left(\frac{5}{10}\right)^2 (-8) \Rightarrow v_1 = 0$$

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